

# Acute myeloid leukemia

≥20% myeloid blasts — or any blast % with a defining genetic lesion · classification and treatment are now genetically driven · every new diagnosis is triaged for the three emergencies before a regimen is chosen

Compiled from the Encyclopedia AML review note and Fellowship Anki deck · ELN 2022 · WHO 2022 / ICC · NCCN. Items marked † extend beyond the existing review note.

## DIAGNOSIS

- ≥20% myeloid blasts in blood or marrow — or any blast % with a defining genetic lesion: t(8;21), inv(16)/t(16;16), PML::RARA, and (WHO 2022) NPM1 or CEBPA-bZIP
- Also AML at <20%: **myeloid sarcoma** / extramedullary disease
- “Blasts” on smear is morphologic — **confirm lineage by flow**
- Workup: marrow aspirate + biopsy with **flow, karyotype + FISH, NGS**; CBC/smear (Auer rods); **coags + fibrinogen + D-dimer**; TLS labs; **baseline echo**; HLA typing early; LP only if CNS symptoms or monocytic, after coags corrected
- **Stat FISH/PCR for PML::RARA** on any suspicion of APL
- † Median age ~68. The three triage questions on every new AML: **Is it APL? Is there leukostasis? Is there tumor lysis?**

## ELN 2022 GENETIC RISK

| Risk         | Representative genetics                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Favorable    | t(8;21) RUNX1::RUNX1T1 · inv(16)/t(16;16) CBFB::MYH11 ( <b>core-binding factor</b> ) · <b>mutated NPM1 without FLT3-ITD</b> · in-frame bZIP CEBPA                                                        |
| Intermediate | NPM1 + FLT3-ITD · wild-type NPM1 + FLT3-ITD · t(9;11) MLLT3::KMT2A · anything not otherwise classified                                                                                                   |
| Adverse      | Complex or monosomal karyotype · -5/del(5q), -7, abn(17p) · <b>TP53</b> · MDS-related genes (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2) · KMT2A rearrangements · <b>t(6;9), inv(3)</b> |

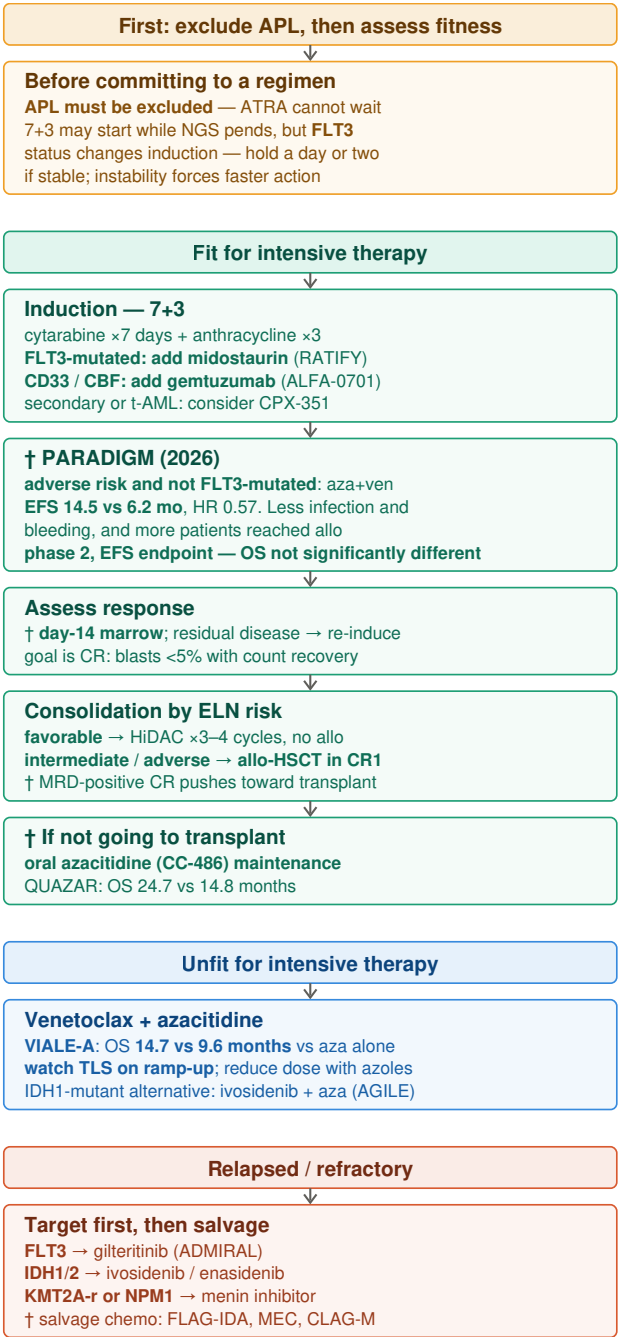
ELN 2022 changes worth knowing: FLT3-ITD is intermediate regardless of allelic ratio (ratio dropped) · only **in-frame bZIP CEBPA** is favorable · **NPM1** with adverse cytogenetics is reclassified adverse · **KIT** co-mutation worsens core-binding-factor AML.

## † GENOTYPE → PHENOTYPE

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| FLT3-ITD      | Hyperleukocytosis; normal karyotype. † <b>CNS risk</b> , most when <b>NPM1 co-mutated</b> |
| KMT2A / 11q23 | Monocytic — gum hypertrophy, leukemia cutis. <b>Strongest CNS predictor</b> (OR ~7.6)     |
| t(8;21)       | <b>Myeloid sarcoma</b> ; aberrant <b>CD19 / CD56</b> by flow                              |
| inv(16)       | <b>Abnormal marrow eosinophils</b> with coarse basophilic granules (FAB M4Eo)             |
| TP53          | Complex karyotype, t-AML; <b>venetoclax-resistant</b> , poor outcome even after allo      |
| DTA           | <b>DNMT3A, TET2, ASXL1 persist in CR</b> as clonal hematopoiesis — <b>not MRD markers</b> |

CNS disease in adult AML is uncommon — ~0.6% at diagnosis, ~3% at relapse; on multivariable analysis **WBC** and **11q23** outweigh FLT3-ITD itself.

## TREATMENT PATHWAY



## REGIMENS & AGENTS

| Regimen                  | Notes                                                                                                                                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7+3                      | <b>Cytarabine 100–200 mg/m<sup>2</sup>/day</b> continuous infusion ×7 days († <b>SHC uses 100</b> ) + <b>daunorubicin 60 mg/m<sup>2</sup> IV days 1–3</b> . E1900 showed <b>90 &gt; 45</b> , but † <b>AML17 found 90 vs 60 gave no CR or OS benefit and doubled 60-day mortality</b> — 60 is the modern standard |
| + midostaurin            | FLT3-mutated (RATIFY). Quizartinib for FLT3-ITD (QuANTUM-First)                                                                                                                                                                                                                                                  |
| + gemtuzumab             | CD33-positive, fractionated dosing; greatest benefit in <b>core-binding factor</b> (ALFA-0701)                                                                                                                                                                                                                   |
| CPX-351                  | Liposomal cytarabine + daunorubicin for t-AML / AML-MRC. <b>Caveat</b> : comparator was 7+3, not ven/aza — niche today is the fitter older adult bridging to allo                                                                                                                                                |
| HiDAC                    | 3 g/m <sup>2</sup> q12h ×3–4 cycles; best in CBF. <b>Age &gt;60 → IDAC 1.5 g/m<sup>2</sup></b> — 3 g/m <sup>2</sup> is inappropriate (ocular, cerebellar, infectious toxicity). Renal dose-reduce                                                                                                                |
| Venetoclax + azacitidine | Unfit patients (VIALE-A). Ramp-up with TLS prophylaxis; <b>reduce venetoclax with CYP3A inhibitors</b>                                                                                                                                                                                                           |
| † Salvage                | FLAG-IDA · MEC · CLAG-M · clinical trial · second transplant                                                                                                                                                                                                                                                     |

## † MRD AND THE TRANSPLANT DECISION

- **Measure MRD at the end of every cycle** and again before transplant — order **HemeSTAMP MRD** (NGS) + **multiparameter flow cytometry**; add **NPM1 PCR** where applicable
- **MRD-positive CR is not CR for planning purposes** — it predicts relapse and moves an otherwise favorable-risk patient toward transplant
- **Allo-HSCT in CR1**: generally **not** for favorable risk · **yes** for intermediate and adverse risk · modified by MRD, age, comorbidity, and donor availability
- Allo cures through **graft-versus-leukemia**; **autologous transplant is not used** — the graft carries the same mutations
- Fitness for intensive therapy is a composite of age, performance status and comorbidity (HCT-CI) — not age alone

THE THREE EMERGENCIES

**APL**

↓

**Recognize**  
DIC + bleeding + hypergranular promyelocytes  
**bundled Auer rods**  
microgranular variant can look monocytic

↓

**Start ATRA immediately**  
do **not** wait for PML::RARA — hemorrhage is the leading cause of APL death, and it is preventable  
**45 mg/m²/day PO divided q12h** (22.5 mg/m² BID);  
↑ **a one-time dose overnight is fine**

↓

**Support aggressively**  
**fibrinogen >100–150** (cryo), **plt >30–50k**, INR normal  
coags q6–12 h; avoid LP and central lines

↓

**Then treat**  
**ATRA + arsenic**, chemo-free, low/int risk  
high risk (WBC >10k): add anthracycline  
cure >90%; **do NOT leukapheresis — worsens DIC**

**Leukostasis**

↓

**Hyperleukocytosis + symptoms**  
myeloid blasts are large and sticky — symptoms at lower counts than ALL/CLL, often >50–100k  
**CNS** (headache, confusion, vision) + **pulmonary** (hypoxia)

↓

↑ **Management — with doses**  
**hydroxyurea up to 2 g PO q6h**, goal **WBC <20k**  
↑ literature 50–100 mg/kg/day; **mucositis ≥75 mg/kg/day**  
HU takes days — if too slow, **cytarabine 1 g ×1**  
**urgent induction is definitive; leukapheresis if severe**  
— **never in APL**, no mortality benefit. **No RBC transfusion**  
hydrate + TLS ppx. ↑ **dex 10 mg IV q6h** if monocytic + ARF

**Tumor lysis**

↓

**Cairo-Bishop**  
lab TLS = ≥2 of ↑K, ↑phosphate, ↑uric acid, ↓Ca  
clinical TLS = lab TLS + AKI, arrhythmia or seizure

↓

**Prevent and treat**  
**aggressive IV hydration** is the cornerstone  
allopurinol 300 mg BID standard risk  
**rasburicase** if high risk or uric acid >8 — **not in G6PD**  
**no urinary alkalization**

DRUG TOXICITY & MONITORING

| Agent                     | Watch for                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cytarabine                | Pyrimidine analog. High-dose: <b>cerebellar toxicity — cerebellar exam before every HiDAC dose, hold if abnormal; conjunctivitis → prophylactic steroid eye drops</b> ; ara-C syndrome (fever, myalgia, rash) |
| Anthracyclines            | Cumulative, dose-dependent <b>cardiotoxicity</b> — track total dose, baseline ± serial LVEF. <b>Vesicant</b> — central access                                                                                 |
| Venetoclax                | <b>Tumor lysis on ramp-up</b> ; neutropenia; <b>dose-reduce with azole antifungals</b> (CYP3A)                                                                                                                |
| Azacitidine vs decitabine | Both hypomethylating; <b>azacitidine incorporates into RNA and DNA</b> , decitabine DNA-only. Azacitidine: injection-site reactions                                                                           |
| Gemtuzumab                | Anti-CD33 calicheamicin ADC — <b>hepatotoxicity and VOD/SOS</b> , especially peri-transplant; fractionated dosing lowers risk                                                                                 |
| Arsenic trioxide          | <b>QT prolongation</b> — monitor ECG, replete K and Mg                                                                                                                                                        |
| Differentiation syndrome  | ATRA/ATO and also IDH and menin inhibitors. Fever, dyspnea, hypoxia, infiltrates, effusions, weight gain, AKI → <b>dexamethasone 10 mg IV BID at first suspicion</b>                                          |
| Menin inhibitors          | Revumenib, ziftomenib — differentiation syndrome + <b>QT prolongation</b>                                                                                                                                     |

TRIAL EVIDENCE

|                  |                                                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RATIFY           | Midostaurin + 7+3 in FLT3-mutated — survival benefit                                                                                                                                                                         |
| QuANTUM-First    | Quizartinib + 7+3 in FLT3-ITD                                                                                                                                                                                                |
| ALFA-0701        | Fractionated gemtuzumab + 7+3 in CD33-positive disease                                                                                                                                                                       |
| E1900            | Daunorubicin <b>90 vs 45 mg/m²</b> — better CR and OS in age <60                                                                                                                                                             |
| ↑ AML17          | Daunorubicin <b>90 vs 60 mg/m² — no CR or OS difference</b> , and 60-day mortality <b>10% vs 5%</b> (HR 1.98). <b>Read with E1900: 60 is standard, not 90</b> (Burnett, Blood 2015)                                          |
| VIALE-A          | Azacitidine + venetoclax vs azacitidine — OS <b>14.7 vs 9.6 months</b> (DiNardo, NEJM 2020)                                                                                                                                  |
| ↑ PARADIGM       | <b>Aza-ven vs intensive induction in fit patients</b> (CBF/FLT3/ NPM1<60 excluded, 72% ELN adverse): EFS <b>14.5 vs 6.2 mo</b> , HR 0.57; less infection and bleeding. <b>Phase 2; OS not significant</b> (Fathi, NEJM 2026) |
| AGILE            | Ivosidenib + azacitidine in IDH1-mutant unfit patients                                                                                                                                                                       |
| ADMIRAL          | Gilteritinib in relapsed FLT3-mutated AML — survival benefit                                                                                                                                                                 |
| APL0406          | ATRA + arsenic beat ATRA + chemo in low/intermediate-risk APL (Lo-Coco, NEJM 2013)                                                                                                                                           |
| ↑ QUAZAR AML-001 | Oral azacitidine (CC-486) maintenance in CR1 — OS <b>24.7 vs 14.8 months</b>                                                                                                                                                 |
| AML15            | IDAC vs HiDAC consolidation — no overall survival difference                                                                                                                                                                 |
| SWOG S1203       | IA regimen showed no benefit and more toxicity — <b>7+3 remains standard</b>                                                                                                                                                 |

GENETICS — FINE POINTS

|                 |                                                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inv(3) / t(3;3) | MECOM (EV11, 3q26) activation; classically <b>normal or high platelets</b> ; adverse and <b>venetoclax-resistant</b>                                                     |
| t(6;9)          | DEK::NUP214 — associates with <b>FLT3-ITD and basophilia</b> ; adverse                                                                                                   |
| NPM1 flow mimic | NPM1-mutated AML can be <b>HLA-DR– / CD34–</b> and mimic APL — confirm PML::RARA                                                                                         |
| Therapy-related | After <b>alkylators/RT</b> : del(5q)/del(7q) ± TP53, latency ~5–7 yr. After <b>topoisomerase-II: 11q23 KMT2A</b> , latency ~2–3 yr                                       |
| ↑ Germline      | <b>DDX41, RUNX1, CEBPA, GATA2</b> , telomere biology — send germline testing if young, family history, or multiple cytopenias; <b>it changes related-donor selection</b> |

Acute myeloid leukemia

† FOUR WAYS AN ACUTE LEUKEMIC DIES IN THE ED

- 1. Leukostasis — CVA, MI, respiratory failure
- 2. Bleeding / DIC / thrombosis
- 3. Infection — very low threshold for broad-spectrum antibiotics
- 4. Tumor lysis — renal failure, electrolyte derangement

OVERNIGHT DECISIONS

- 1. Triage floor vs ICU — leukapheresis and CRRT both require ICU
- 2. Rule out APL — start ATRA on any suspicion, do not wait for PML::RARA
- 3. Cyto reduction — see doses at right
- 4. TLS — start fluids + allopurinol, set lab frequency (q4h to daily). Rasburicase if uric acid >8: 0.15–0.2 mg/kg IV, in adults usually a single dose capped at 6 mg (label allows 0.2 mg/kg daily ×5 days). Check G6PD — contraindicated if deficient

SUPPORTIVE CARE

- Transfusion: platelets >10k (>20–50k if bleeding, procedure, or APL); irradiated and leukoreduced products
- † Prophylaxis during induction: levofloxacin 500 mg PO daily · posaconazole DR 300 mg BID ×1 day, then 300 mg daily · acyclovir 400 mg PO BID (HSV-seropositive)
- † Acyclovir vs valganciclovir: acyclovir covers HSV/VZV only; valganciclovir is a CMV drug with no role in standard AML induction — CMV reactivation is a transplant / T-cell-depletion problem (allo-HSCT, ATG, alemtuzumab). In CMV-seropositive allo-HSCT, letermovir has replaced valganciclovir for primary prophylaxis (valganciclovir is myelosuppressive), and because letermovir has no HSV/VZV activity, acyclovir continues alongside it. Valganciclovir today is mostly preemptive therapy for CMV DNAemia
- † Posaconazole is a strong CYP3A4 inhibitor — that is what forces the venetoclax reduction: 400 mg → 70 mg daily with posaconazole (100 mg with other strong inhibitors such as voriconazole); during ramp-up and beyond
- † Hepatitis B: screen HBsAg + anti-HBc before therapy. HBsAg-positive → entecavir or tenofovir (preferred over lamivudine) for any systemic anticancer therapy, continued ≥12 months past the last cycle. Isolated anti-HBc-positive → prophylax or monitor HBV DNA closely; AML chemotherapy carries reactivation risk similar to lymphoma
- Febrile neutropenia: fever ≥38.3°C once (or ≥38.0°C for 1 h) with ANC <500 → cefepime 2 g IV q8h within 1 hour. Add vancomycin for hemodynamic instability, line infection, skin/soft tissue infection, pneumonia or known MRSA. † Severe mucositis is a softer trigger than it looks — IDSA lists it only when fluoroquinolone prophylaxis was used and ceftazidime is the empiric agent (viridans strep); cefepime already covers viridans strep. Antifungal if fever persists >4–7 days
- Fertility counseling before chemotherapy; central venous access; early palliative care in older or adverse-risk disease

WHAT TO SEND

| Test               | When                                                                                                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peripheral studies | Flow cytometry, AML prognosis assay (NPM1, FLT3-ITD, FLT3-TKD), IDH mutation analysis. Send if WBC >10k and marrow will be delayed, or the patient is old enough that you would rather not subject them to a marrow |
| Cytogenetics       | FISH for PML::RARA and BCR::ABL1. Send from peripheral blood if there is any concern for APL or a delay in getting marrow                                                                                           |
| HLA typing         | Send if the patient is a possible allogeneic transplant candidate                                                                                                                                                   |

CYTOREDUCTION OVERNIGHT

| Dose and trigger |                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydroxyurea      | Consider if WBC >20k. Overnight you can order a one-time dose. Typical range 500 mg–2 g, max 2 g q6h. † Literature range 50–100 mg/kg/day; grade 3–4 mucositis rises at ≥75 mg/kg/day, so 2 g q6h is the ceiling, not the starting dose                                                                                                                   |
| Leukapheresis    | Consider for evidence of leukostasis — headache, blurry vision, dyspnea/hypoxia, chest pain — typically with blasts >50k in AML (rare in ALL). Page apheresis 12027. Ask ER or the MICU triage fellow for a trialysis line; admit to MICU. Ask the apheresis RN to save the product in the FGR fridge and coordinate with the research team. Never in APL |
| Cytarabine       | 1 g ×1 may be considered for leukostasis when apheresis will be delayed                                                                                                                                                                                                                                                                                   |

APL OVERNIGHT

- Start ATRA on any concern for APL — do not wait for PML::RARA
- Smear: bilobed / folded nuclei ± Auer rods. Page pathology on call (12005) if unsure
- ATRA 45 mg/m²/day PO divided q12h (= 22.5 mg/m² BID) — overnight you can give a one-time dose
- DIC labs q6–8h. Transfuse platelets >30, fibrinogen >100, INR <2

INDUCTION REGIMENS — DOSING

| Regimen           | Dosing                                                                                                                                                                                                            |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7+3               | Cytarabine 100–200 mg/m²/day IV continuous infusion ×7 days from day 1 († SHC uses 100) + daunorubicin 60 mg/m² IV daily days 1–3 AML17 / Burnett 2015                                                            |
| 7+3 + GO          | Add gemtuzumab ozogamicin 3 mg/m² (max 4.5 mg) IV over 2 h on days 1, 4, 7 ALFA-0701 / Lambert 2019                                                                                                               |
| 7+3 + midostaurin | Add midostaurin 50 mg PO q12h on days 8–21 RATIFY / Stone 2017                                                                                                                                                    |
| CPX-351           | Liposomal daunorubicin 44 mg/m² + cytarabine 100 mg/m² IV on days 1, 3, 5 Lancel 2018                                                                                                                             |
| FLAG-IDA-VEN      | Fludarabine 30 mg/m² + cytarabine 1.5–2 g/m² IV days 2–6; idarubicin IV (ND-AML 8 mg/m² days 4–6; R/R 6 mg/m² days 4–5); filgrastim 5 mcg/kg days 1–7; venetoclax ramp-up (100/200/400 mg) days 1–14 DiNardo 2021 |
| CLAD / LDAC / VEN | Cladribine 5 mg/m² IV days 1–5 + cytarabine 20 mg/m² (SQ BID days 1–10; † 20 mg/m² IV daily preferred at SHC) + venetoclax PO days 1–21, alternating with azacitidine Kadia 2022                                  |
| HMA / VEN         | Venetoclax (ramp-up) daily ×28 days + decitabine 20 mg/m² days 1–5 or azacitidine 75 mg/m² days 1–7 VIALE-A / DiNardo 2020                                                                                        |

APL INDUCTION — DOSING

| Regimen          | Dosing                                                                                                                                                                                          |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATO / ATRA       | Arsenic trioxide 0.15 mg/kg IV over 2 h daily from day 1 + ATRA 22.5 mg/m² PO BID from day 1 — both until hematologic CR or max 60 days APL0406 / Lo-Coco 2013                                  |
| ATO / ATRA / IDA | ATO 0.15 mg/kg IV days 9–36 + ATRA 22.5 mg/m² BID days 1–36 + idarubicin on days 2, 4, 6, 8 by age: ≤61 yr 12 mg/m² · 61–70 yr 9 mg/m² · >70 yr 6 mg/m² APML4 / Iland 2012                      |
| ATO / ATRA / GO  | ATO 0.3 mg/kg IV daily days 1–5, then 0.25 mg/kg twice weekly weeks 2–8 + ATRA 22.5 mg/m² BID from day 1 until CR or max 60 days + gemtuzumab 6 mg/m² IV once on day 1 AML17 APL / Burnett 2015 |